Approach to a Patient with Thyroid Disease

Sanjay Jumani, MD

Clinical Assistant Professor of Medicine

Division of Endocrinology, Diabetes, and Metabolic Diseases

Disclosures

- I have nothing to disclose

Objectives

By exam time, you should be able to do the following:

- Distinguish the clinical signs of hyper- and hypothyroidism and interpret thyroid hormone lab assays.
- Identify etiologies of hyper- and hypothyroidism.
- Generate the treatment options for hyper- and hypothyroidism.
- Formulate the appropriate evaluation of a thyroid nodule and recognize the different types of thyroid cancer.

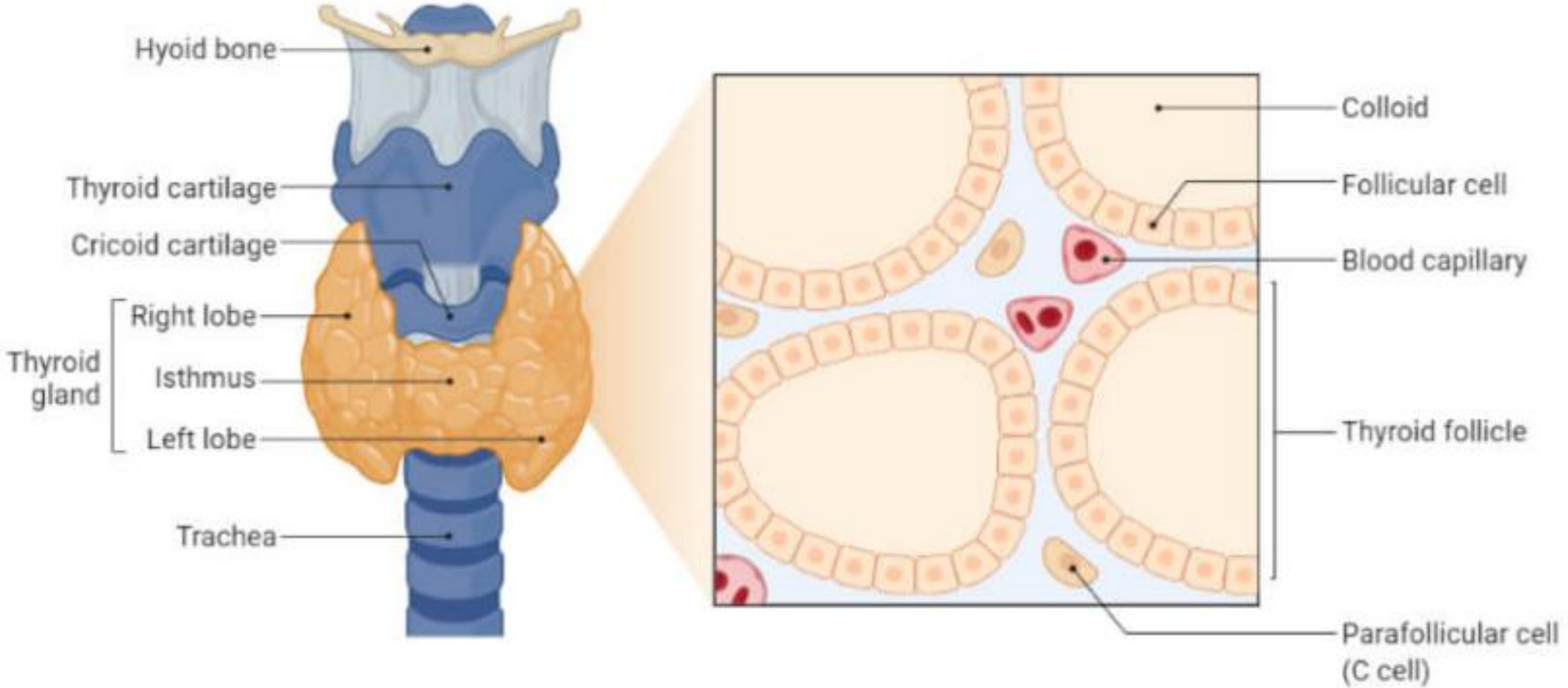
Key

- *** Clinical Pearl
- *** Sometimes on Step 1

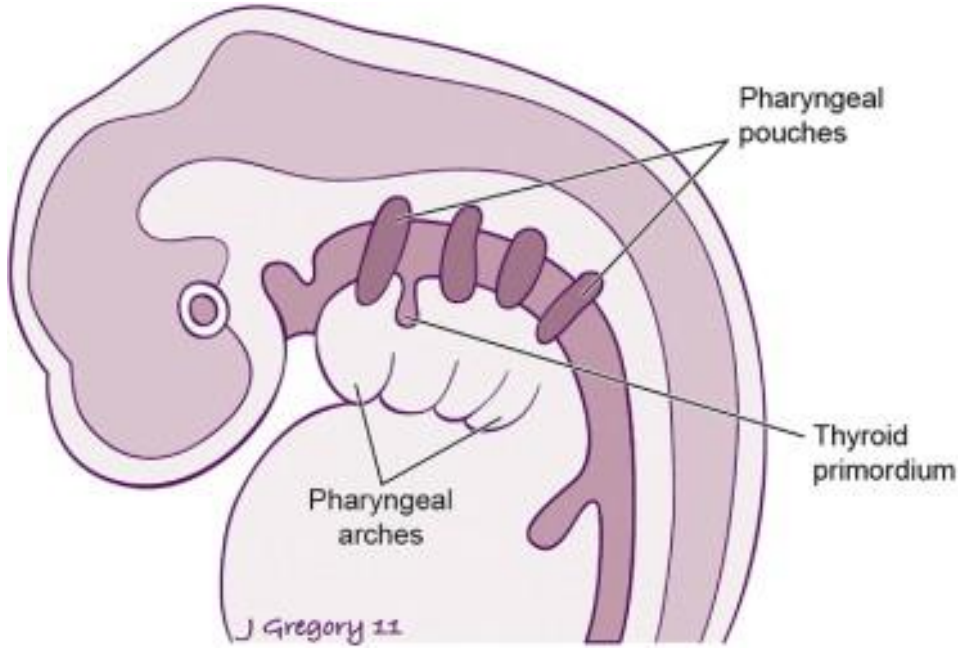
Thyroid Basics

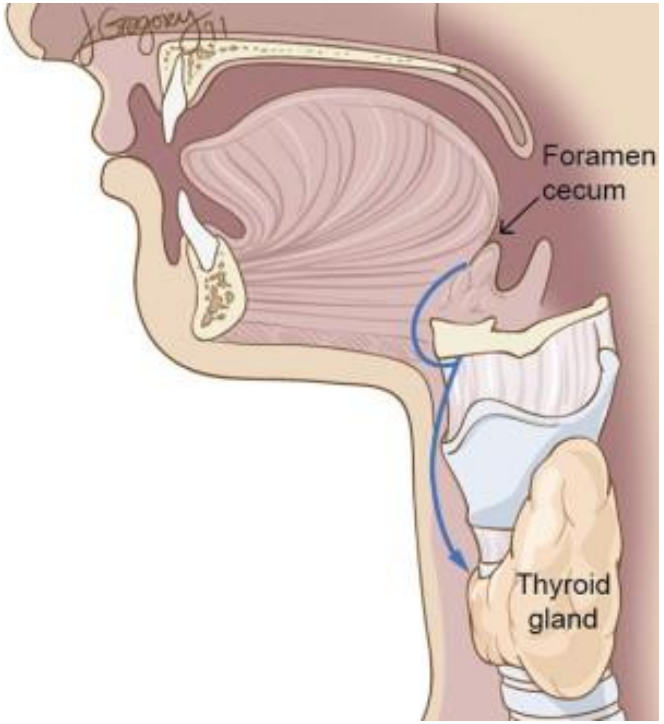
Thyroid Anatomy, Embryology, Hormone Production, and Kinetics

The Thyroid


Hyoid bone
Thyroid cartilage
Cricoid cartilage
Thyroid gland
Right lobe
Isthmus
Left lobe
Trachea
Colloid
Follicular cell
Blood capillary
Thyroid follicle
Parafollicular cell (C cell)

Brief Embryology Review



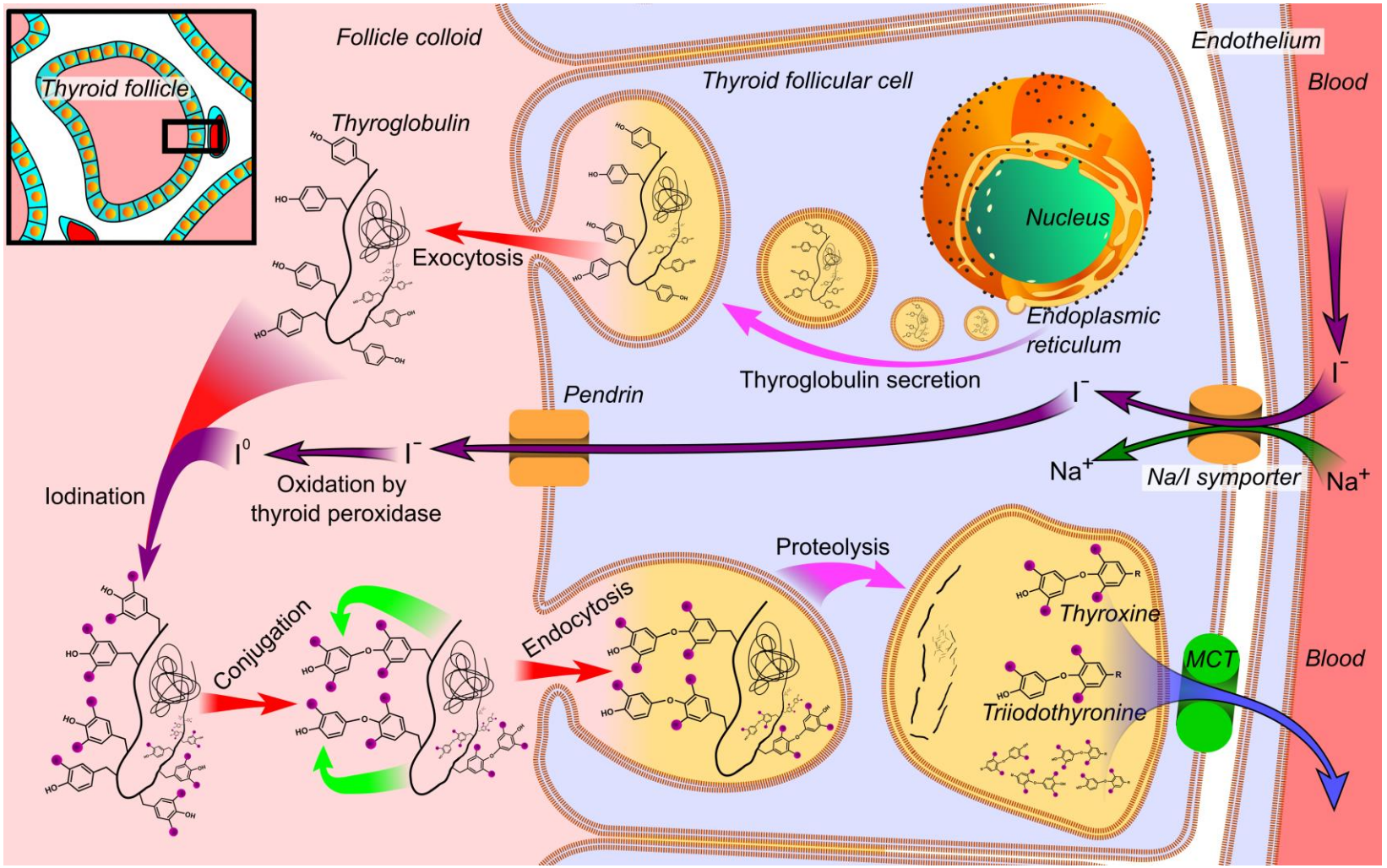






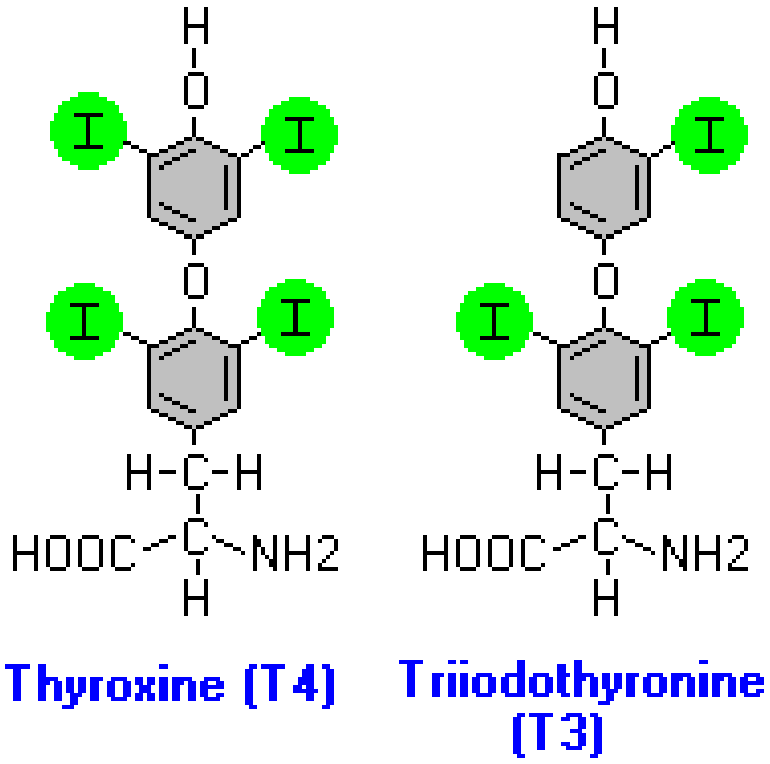
★★★ Thyroglossal Duct Cyst

Thyroid Hormone Synthesis



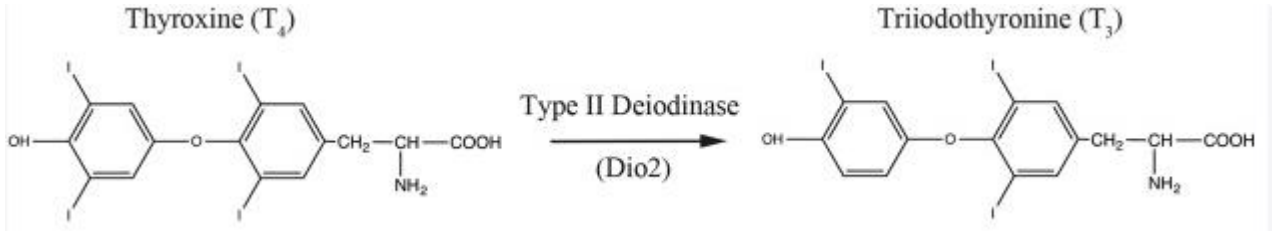
Thyroid Hormone Kinetics

- Triiodothyronine (T3) – active thyroid hormone (this is the one that does the stuff that thyroid hormone does). Dissipates in Minutes/hours
  - Three iodine atoms
- Thyroxine (T4) – transport form, stable in the serum for days (the major circulating form)
  - Four iodine atoms



Thyroid Hormone Kinetics

- Thyroid secretes T3 and T4 in a 1:13 ratio (most of it is the shelf stable form)
- In the tissues, deiodinase enzymes cleave the last Iodine atom and “activate” thyroid hormone by converting from T4 → T3



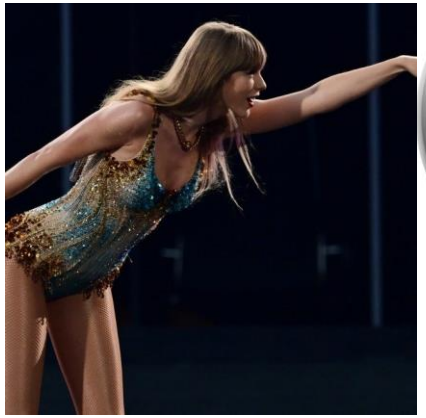
Thyroid Hormone Kinetics – Bottom Line

- Thyroid hormone is released by the thyroid
- Thyroid circulates in the blood stream bound to proteins. Bound thyroid hormone is inert.
  - Thyroid binding globulin
  - Albumin
- In the tissues, it gets “activated” by deiodinases in tissues
- Labs can measure
  - “Free T4” – unbound, active thyroid hormone
  - “Total T4” – Unbound + Bound thyroid hormone
- *** The most stable measurement of active thyroid **hormone** is “Free T4”
- *** The most stable measurement of active thyroid **function** is “TSH”… coming soon

Thyroid Hormone Physiology

The “HPT” Axis

An Endocrine Paradigm - Physiology









Too Cold

Too Hot

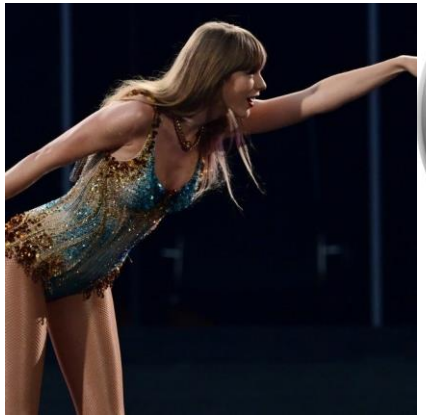
An Endocrine Paradigm - Pathology

Is the house responding appropriately?



Is the Nest communicating with my furnace?

Is the next working?





Did I not turn on the Nest?

Did I even wake up this morning?



Is my furnace working?



Brr...

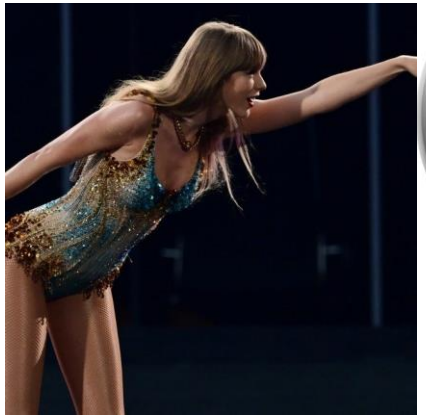
An Endocrine Paradigm – Pathology



Is another nest telling my furnace to work even though it told her to stop?

Is my house on fire?





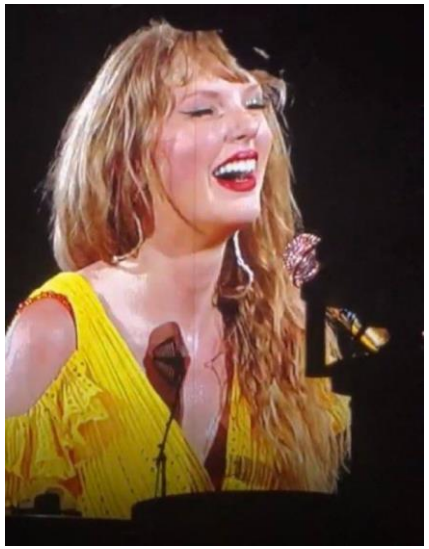


Is the nest on auto mode?

Did I set the Nest too high?



Is the furnace haunted?





“Central”

“Tertiary”



“Secondary”



“Primary”



“Resistance”





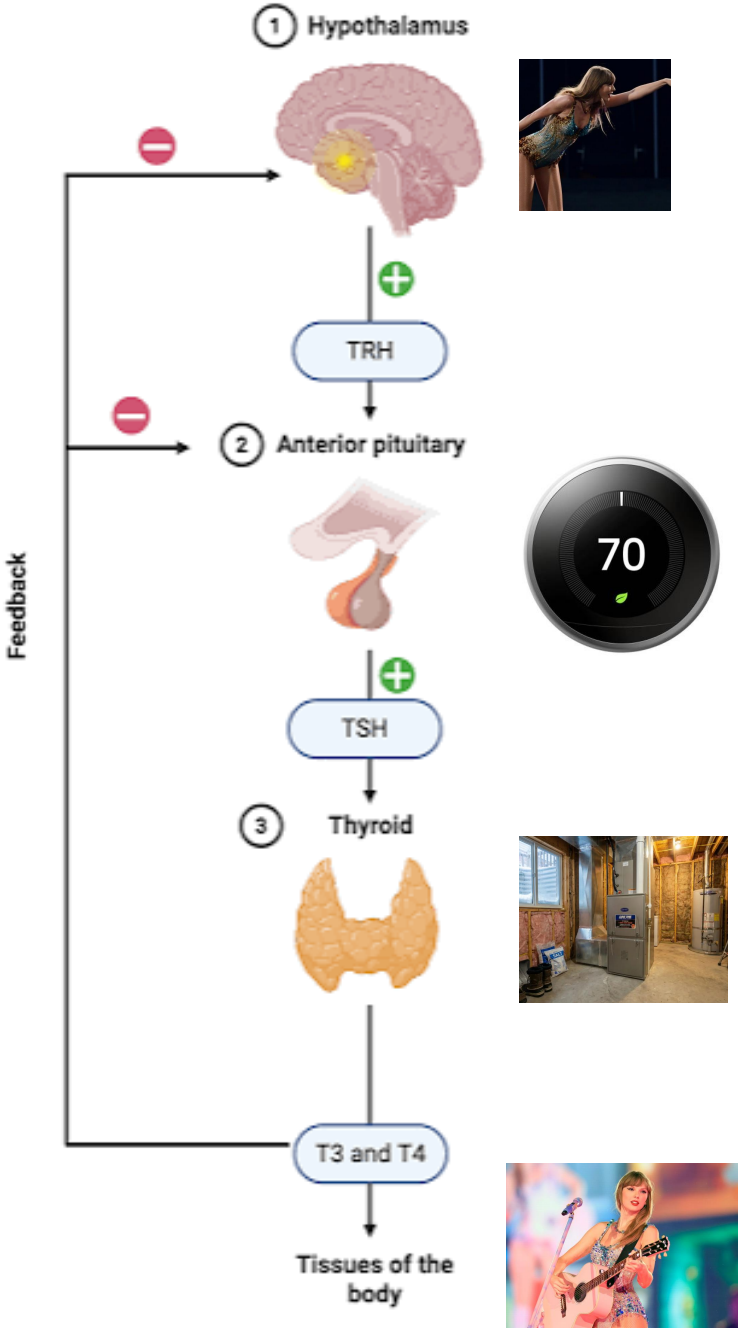






“Ectopic”

The HPT Axis


1 Hypothalamus
TRH
2 Anterior pituitary
TSH
3 Thyroid
T3 and T4
Tissues of the body
Feedback

70

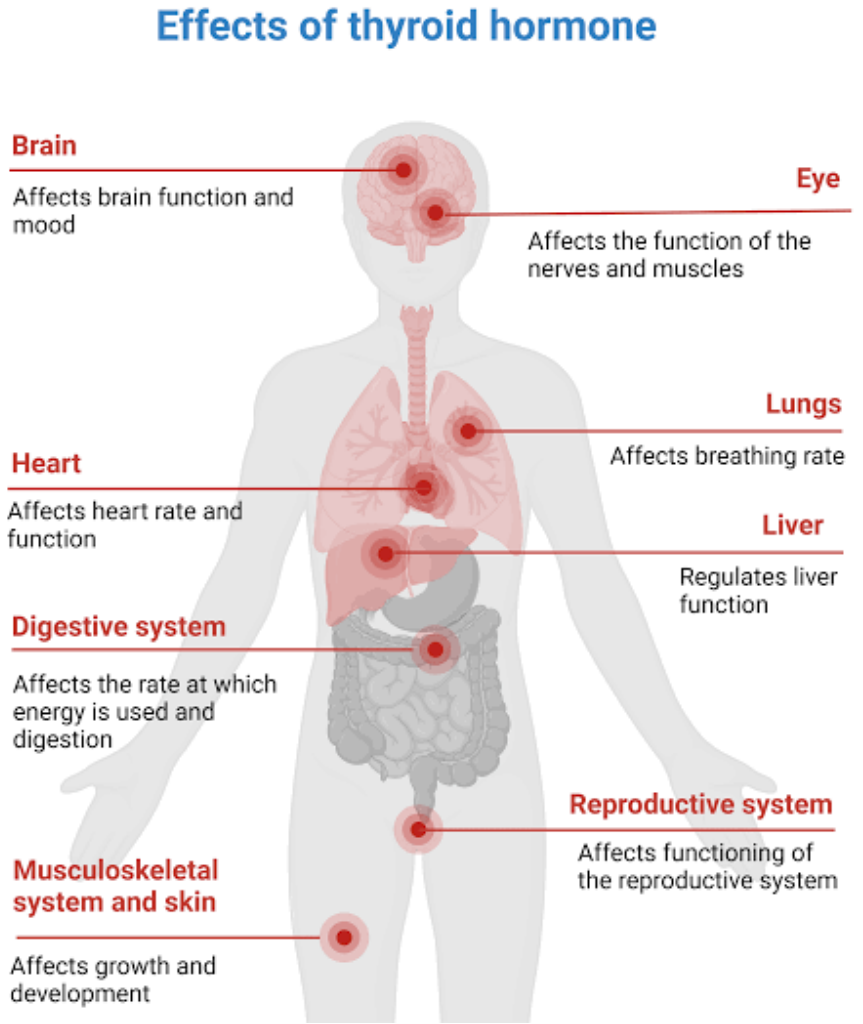
Laboratory Evaluation

- Small Changes in serum Free T4 induce very large, measurable changes in serum TSH
- As a result, thyroid **function** is **best assessed by measuring serum TSH**
- **In most cases you will start evaluation by checking a TSH.**
  - **An Abnormal TSH should prompt you to check Free T4 and sometimes Total T3**

Thyroid Hormone Function

What does thyroid hormone actually do?

- Thyroid hormone has a nuclear receptor
  - Thyroid hormone must enter the cell to bind its receptor
  - Activated thyroid receptor goes into the nucleus and binds DNA
  - This binding affects **gene expression and activation**
- Thyroid hormone thus does not work quickly



Think of Thyroid Hormone as your “metabolism” hormone

- Temperature Control
- Energy expenditure (weight, muscle function)
- Cell turnover (think skin, nails, hair, gut)
- Heart rate
- Respiratory rate
- Digestive system
- Liver function (cholesterol metabolism)
- Reproductive function
- Mood/Cognition
- Growth

Case 1

- A 34-year-old cis woman presents with 3 months of weight loss, palpitations, insomnia, heat intolerance and increased stool frequency. She notes her menses have become lighter and more infrequent.
- Her vital signs are remarkable for a mildly elevated systolic blood pressure with a pulse of 112 bpm.

What do you suspect?

- Hypothyroidism
- Hyperthyroidism

Hyperthyroidism

- General
  - Weight Loss
  - Sweating and heat intolerance
- Cardiovascular
  - Palpitations
  - Tachycardia
- Musculoskeletal
  - Proximal muscle weakness
  - Osteoporosis, increased fracture risk
- Reproductive
  - Oligo-/Amenorrhea
- GI
  - Hyperdefecation (not diarrhea)
- Neuropsych
  - Nervousness
  - Agitation/anxiety
  - Insomnia
  - Tremor

Physical Exam

- Vitals: Sins tachycardia, Atrial fibrillation
- Neck: Goiter, thyroid bruit
- Skin: Warm, moist skin (vasodilation)
  - Onycholysis (loosening of nails from the nail bed)
  - Pretibial myxedema
- Neuro: Brisk reflexes, Tremor





Causes of hyperthyroidism

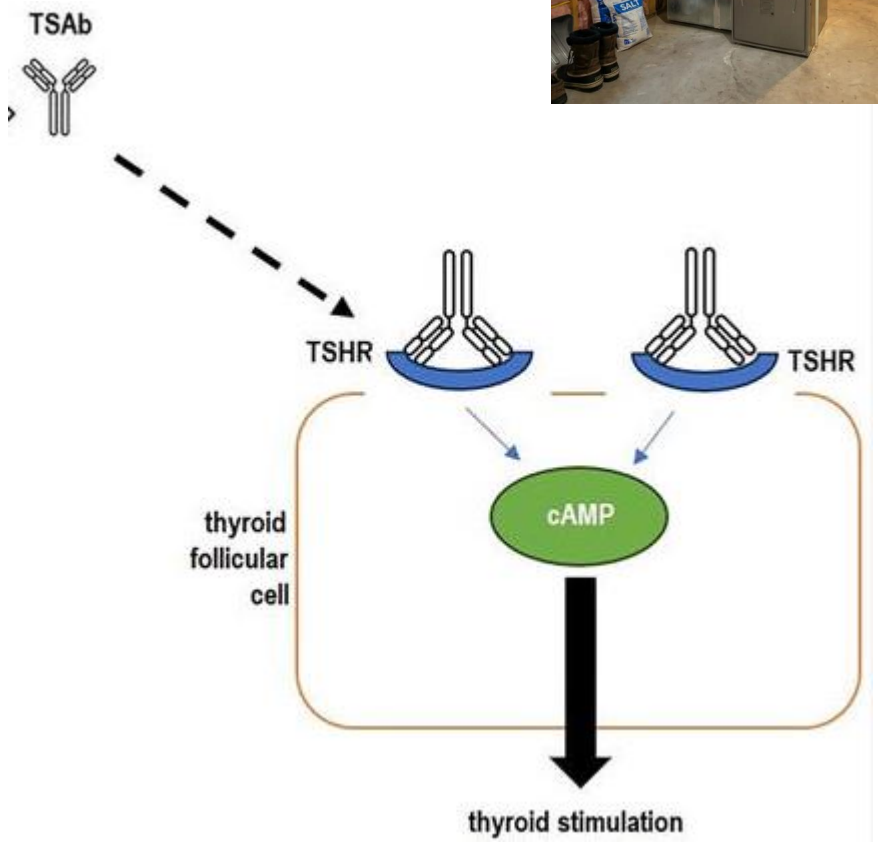
- **Primary Hyperthyroidism**
  - **Graves disease**
  - **Toxic adenoma/Toxic multinodular goiter**
  - **Thyroiditis**
  - Congenital
- Central Hyperthyroidism
  - TSH producing adenoma
- Ectopic Hyperthyroidism
  - Multiple Gestation/trophoblastic disease
  - Struma ovarii
- Exogenous thyroid hormone intake

Graves Disease – Primary Hyperthyroidism

- Primary problem: Autoimmune condition due to the presence of thyroid stimulating immunoglobulins
- Activation of the entire thyroid gland



TSH	Free T4	Total T3	TSI (TRAb)



Graves Disease

- Pathognomonic features:
  - Ophthalmopathy (exophthalmos and periorbital myxedema)
  - Pretibial Myxedema
  - Thyroid bruit
  - Acropachy
- TSI is an IgG, it can cross the placenta and lead to transient neonatal Graves disease.



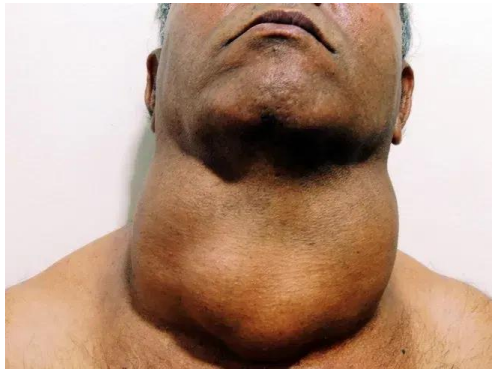




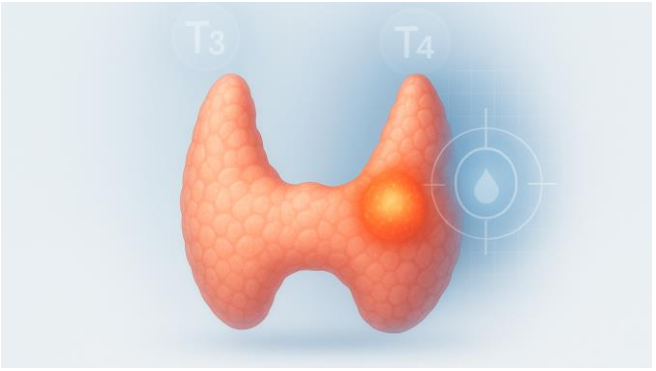
Toxic Adenoma/Toxic Multinodular Goiter



- Primary problem: Adenoma(s) in the thyroid lead to autonomous thyroid hormone secretion
- Nodule produces excess hormones, so the remainder of the thyroid sort of quiets down.



TSH	Free T4	Total T3	TSI (TRAb)
[redacted]	[redacted]	[redacted]	[redacted]e



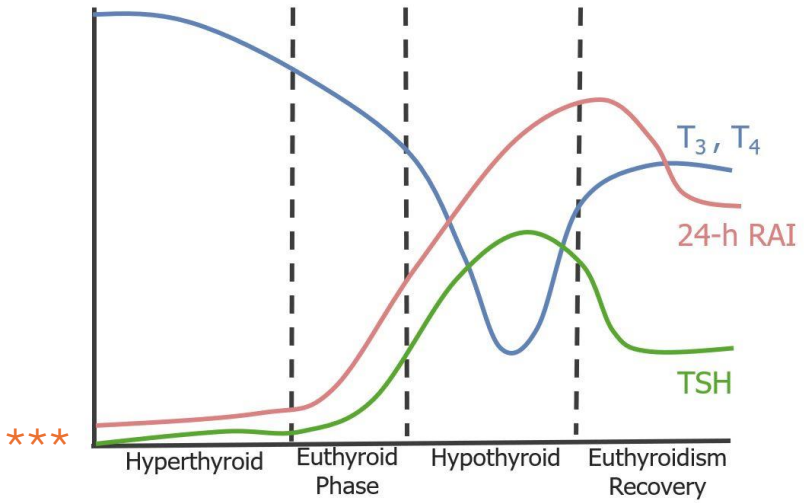
Subacute Thyroiditis “DeQuervain’s”

- Primary problem: inflammation of the thyroid gland, causing thyroid hormone to leak into the circulation.
- Usually associated with pain and tenderness in the neck
- Flu like signs (fatigue, low-grade fever, muscle aches)
- Because there is a lot of thyroid hormone already around, the entire thyroid quiets down

TSH	Free T4	Total T3	TSI (TRAb)
[redacted]	[redacted]	[redacted]	[redacted]

Other forms of thyroiditis associated with hyperthyroidism

- “Silent” Thyroiditis
- Suppurative thyroiditis (infection in the thyroid itself)
- Postpartum thyroiditis
- Drug-induced thyroiditis (amiodarone, immune checkpoint inhibitors [pembrolizumab, ipilimumab], interferons, lithium



Clinical Pearl

- Subacute Thyroiditis usually does not need intense treatment
- Generally resolves spontaneously
- Supportive care: Analgesics (NSAIDs), Glucocorticoids

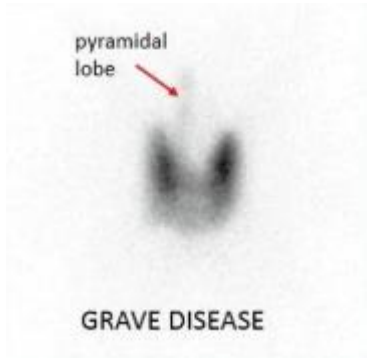
Primary Hyperthyroidism Differential

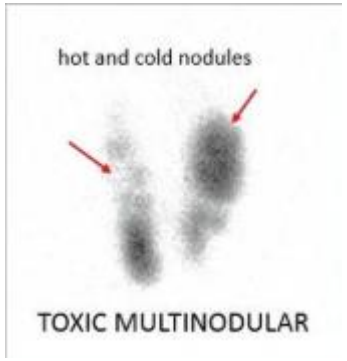
TSH	Free T4	Total T3
↓	↑	↑

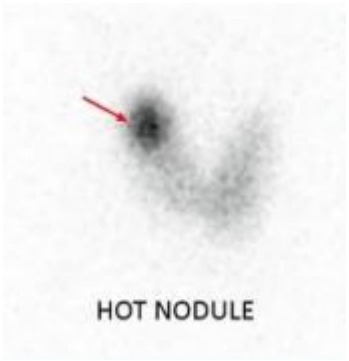
- How can you differentiate between Graves, subacute or painless thyroiditis, or toxic adenoma?
- Can measure Graves antibody (TSI or TRAb)
- Perform Uptake and Scan

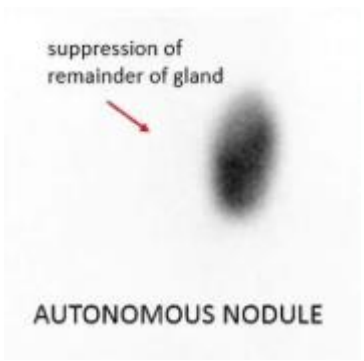
Uptake and Scan

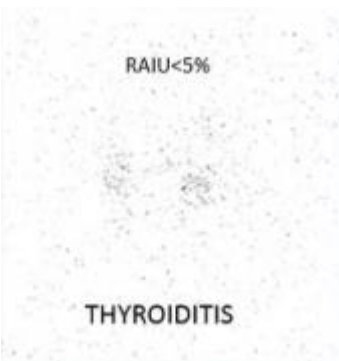
- A small amount of radioactive iodine or technetium-99 is administered and a scan is performed
- High iodine uptake indicates de novo synthesis of hormone
- Hyperactive thyroid tissue will glow











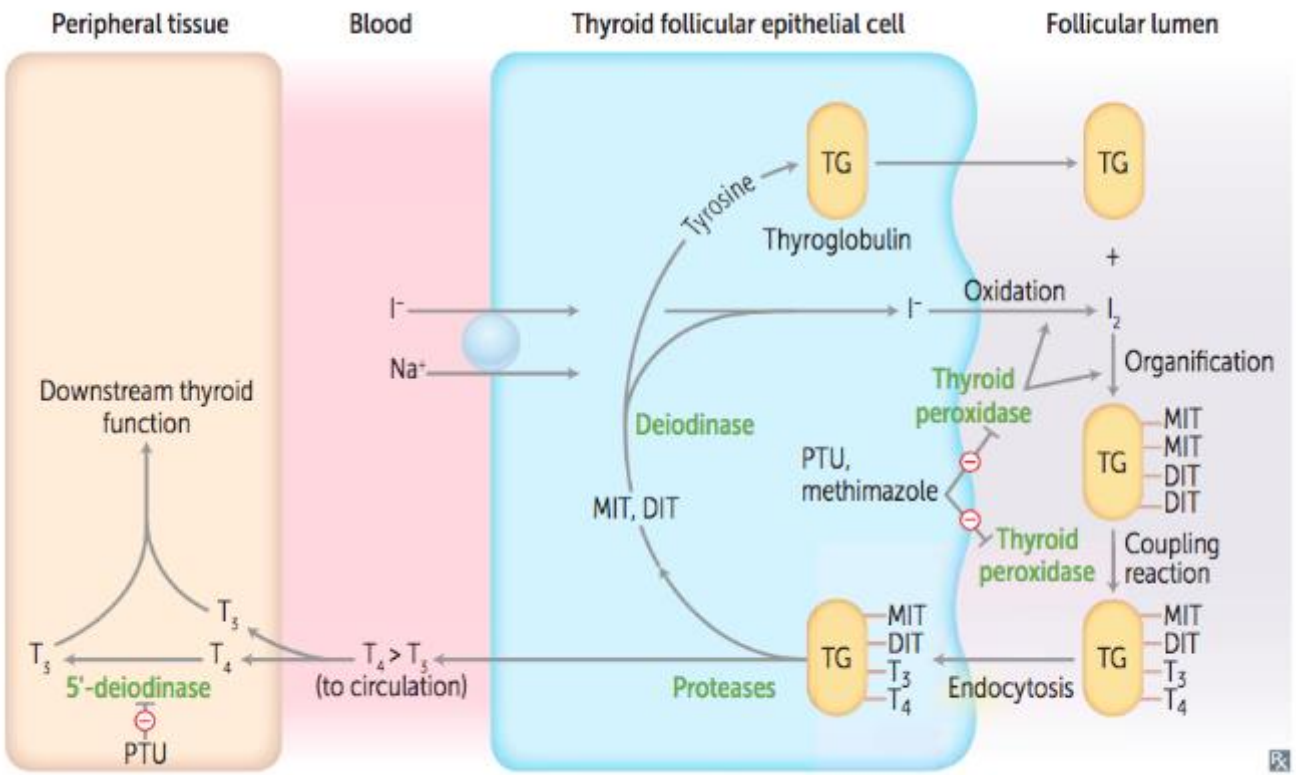
Management of Primary Hyperthyroidism

- Beta Blockers
- Thionamides (Methimazole and Propylthiouracil)
- Radioactive Iodine
- Surgery (hemi- or total thyroidectomy)

Beta Blockers

- Controls the symptoms of hyperthyroidism that are caused by increased beta-adrenergic tone (e.g palpitations, tremors, anxiety, tachycardia)

Thionamides



- Methimazole
- Propylthiouracil (PTU)
- Inhibit organification of iodine to tyrosine and prevent formation of thyroid hormone

Thionamides

- Common: Pruritus, arthalgias, nausea (temporary, respond to diphenhydramine)
- Hepatotoxicity
  - PTU > Methimazole
    - PTU can lead to fulminant hepatic necrosis
- Teratogenicity
  - Methimazole > PTU
    - Methimazole: Aplasia cutis, Tracheoesophageal fistulae
- Agranulocytosis – super rare!
  - Usually within first 3 months of treatment
  - Should think about this if someone develops febrile illness or pharyngitis while on these meds – get a CBC!

Thionamides

- Pretreatment Evaluation: WBC count and Hepatic Function Panel
- Dose based on severity of hyperthyroidism
- Repeat TFTs every 4-6 weeks until on a stable dose
  - Monitor TSH, Free T4, and Total T3 for response
  - TSH takes months to correct after Free T4 and T3 are normal
- Thionamides are continued for 2-9 years (50% remission rate, depending on antibody status
- Long term monitoring for Recurrence

Thyroidectomy

- Definitive therapy for Graves’
- Helpful if singular toxic adenoma
- Indications:
  - Large obstructive goiter
  - Allergic to thionamides/severe AE
  - Unable to receive radioactive iodine
- Requires lifelong hormone replacement therapy after surgery

Radioactive Iodine Ablation

- Administer a pill of radioactive iodine to induce thyroid destruction
- 70% effective (30% need repeat RAI administration)
- Need lifelong thyroid hormone replacement after ablation

What is this pattern?

TSH	Free T4	Total T3
↑	↑	↑





- Central Hyperthyroidism (TSH producing Tumor)

TSH	Free T4	Total T3
Normal	↑	↑

Rapid Fire – Ectopic/Exogenous

	TSH	Free T4	Total T3
Struma Ovarii	↓	↑	↑
Exogenous Thyroid Hormone	↓	↑/normal	↑/normal
Hyper-emesis/ Tropho-blastic disease	↓	↑	↑

- Struma Ovarii: Ectopic Thyroid tissue in an ovarian teratoma
- Exogenous thyroid hormone: Dose of thyroid hormone replacement is too high.
- Trophoblastic disease/Twin Gestation: high levels of bHCG can stimulate the TSH receptor





Thyroid Storm – An Endocrine Emergency! ***

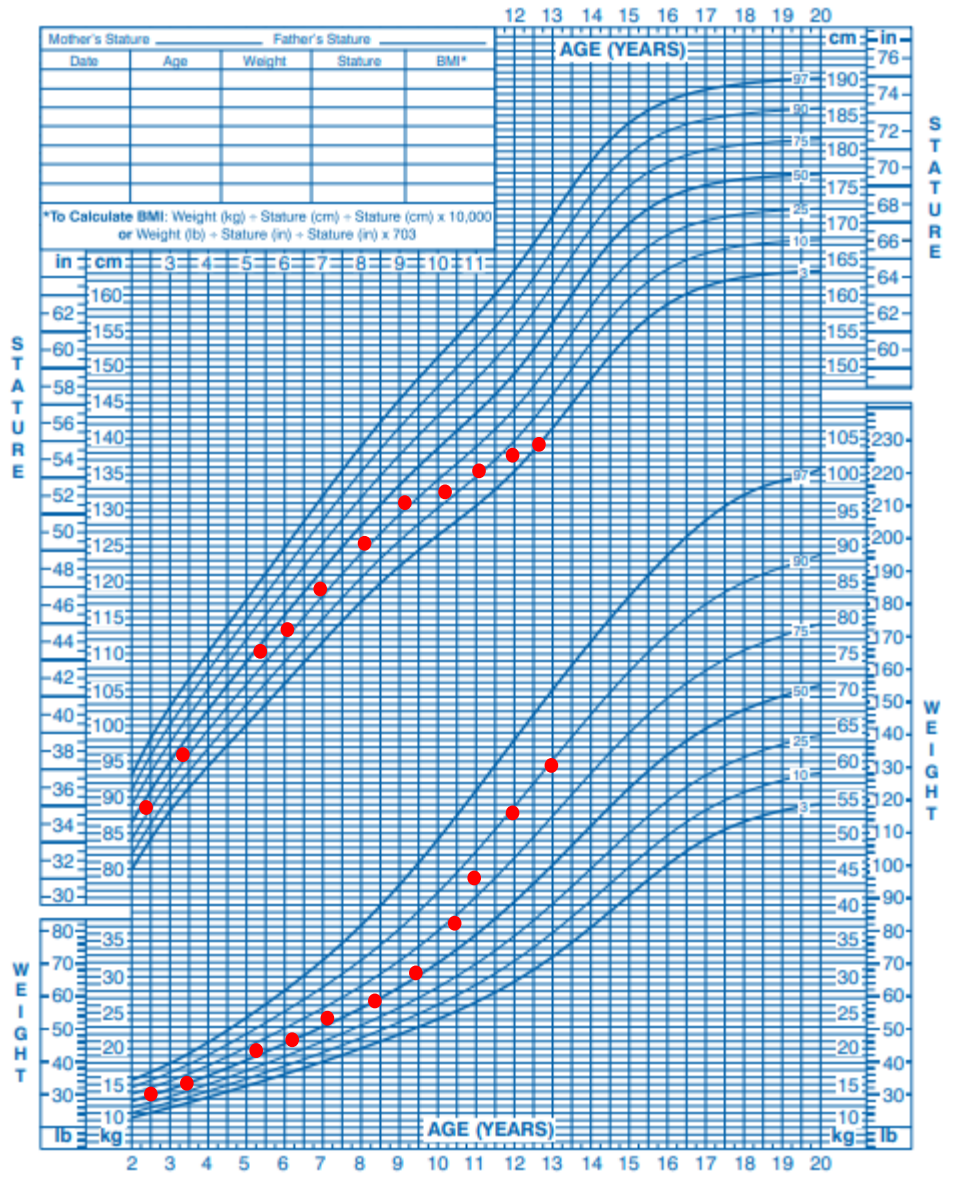
- Rare, life-threatening condition characterized by severe thyrotoxicosis
- Typically in the setting of untreated hyperthyroidism
- Acute stress (infection, surgery, trauma)
- Altered mental status, fever, diarrhea, tachycardia, CHF
- Treatment
  - Beta blocker to control the symptoms
  - Thionamide
  - Iodine solution (potassium iodide) to block the release of thyroid hormone
  - Glucocorticoids to block the deiodinases, promote vasomotor stability

Hyperthyroidism Summary

- Clinical suspicion (signs/symptoms)
  - TSH, then Free T4 and T3
  - Antibodies
  - Uptake and Scan
- All types – beta blockers for symptom control
- Further treatment depends on underlying etiology

Case 2

- A 13 year old cis boy presents with symptoms of severe fatigue, constipation, cold intolerance, and brain fog for the past few years. His growth chart is shown.

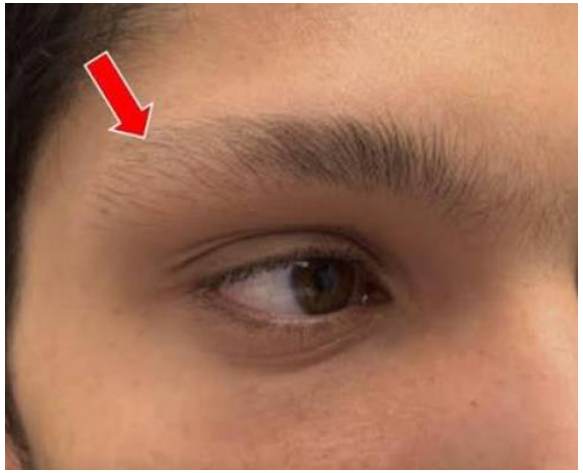


Hypothyroidism

- General
  - Weight gain
  - cold intolerance
  - Lethargy and fatigue
- Cardiovascular
  - Bradycardia
- Musculoskeletal
  - Arthralgia
  - Poor growth
- Neuro
  - Paresthesia
- Reproductive
  - Menstrual disturbance
- GI
  - Constipation
  - Decreased appetite
- Psych
  - Depression
  - Sleepiness
  - Cognitive Impairment

Hypothyroidism - Signs

- General:
  - Slow movements and speech
- Cardiovascular:
  - Diastolic hypertension
  - Bradycardia
- Skin:
  - Dry skin
  - Brittle nails
  - Nonpitting edema (myxedema)
- Neck:
  - Hoarseness
  - Goiter
- Neuro:
  - Slow/absent reflexes





Myxedema

- Generalized nonpitting edema
- Due to glycosaminoglycans in interstitial spaces leading to increased osmotic pressure and water retention

Causes of Hypothyroidism

- Primary Hypothyroidism
  - Hashimoto's Thyroiditis – most common in US
  - Iodine deficiency – most common worldwide
  - Infiltrative disease (e.g. hemochromatosis)
  - Congenital
  - Thyroidectomy or Radioactive iodine
  - External beam irradiation
- Central Hypothyroidism
  - Pituitary issues (adenoma, craniopharyngioma)
- Resistance
  - Genetic resistance to thyroid hormone – we won't cover this, never tested on USMLE

Hashimoto Thyroiditis

- Most common cause of hypothyroidism in US
- Autoimmune destruction of the thyroid gland
- Autoimmune Polyglandular Syndrome ***
  - Defect in AIRE gene
  - Associated with numerous autoimmune endocrinopathies (autoimmune hypothyroidism, Type 1 DM, Addison Disease (adrenal insufficiency), premature ovarian insufficiency)

Diagnosis – Primary Hypothyroidism

- Hashimoto's can be detected by measuring thyroid antibodies
  - Thyroid Peroxidase (TPO) Ab
  - Thyroglobulin Antibody (TgAb)

TSH	Free T4
↑	↓

Diagnosis – Central Hypothyroidism

- Need to have a high clinical suspicion
- Usually screen in people with pituitary surgery

TSH	Free T4
↓	↓

Treatment

- It’s easy-peasy
- Levothyroxine (synthetic T4)
- Dosing: 1.6 mcg/kg/day. Adjust dose TSH and Free T4 every 6-8 weeks until TSH is normalized.
  - TSH continues to be high – increase levothyroxine
  - TSH becomes suppressed – decrease levothyroxine
- TSH is monitored to assess the adequacy of replacement in primary hypothyroidism.
  - A normal TSH means the disease is well controlled, regardless of what the FT4 is
  - A Free T4 is monitored in patients with central hypothyroidism

Levothyroxine

- Several Brands ***
  - Synthroid – the OG, some people swear by it
  - Levoxyl – Gluten Free
  - Tirosint- liquid drops
  - Generic – usually gets the job done, cheap!

Myxedema Coma ***

- Medical Emergency with a high mortality rate
- Severe, longstanding hypothyroidism
- Precipitated by an acute event in a poorly controlled hypothyroid patient (infection, MI, cold exposure, surgery, iodinated contrast)
- Decreased mental status, hypothermia, hypotension, bradycardia, hyponatremia, hypoglycemia, ascites/pleural effusion
- Treatment:
  - Thyroid hormone replacement
  - Glucocorticoids (until the possibility of co-existing adrenal insufficiency has been excluded)
  - Supportive measures

Hypothyroidism Review

- Clinical Suspicion
- Check TSH → if abnormal check Free T4
- Consider TPO Ab or TgAb to evaluate for Hashimoto’s
- Treatment: Thyroid hormone replacement with levothyroxine

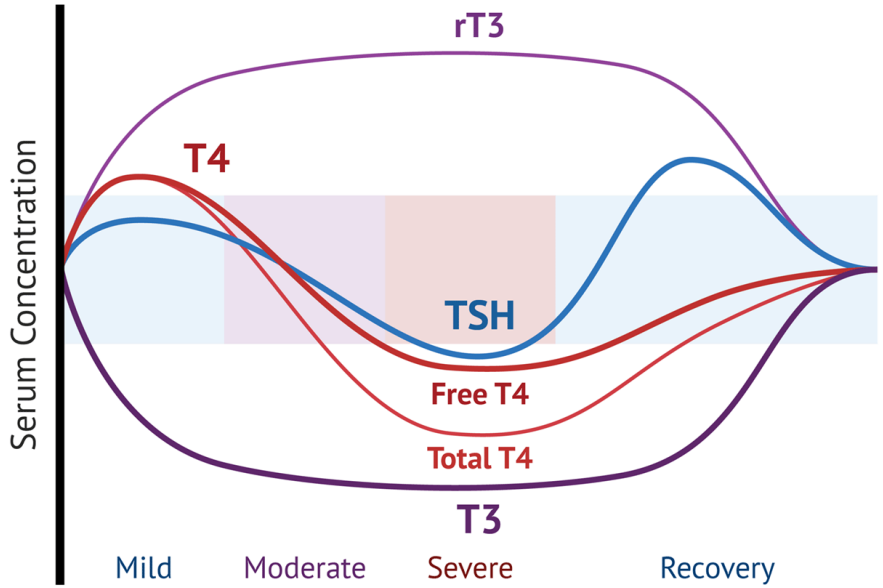
Case 3

- A 76 year old cis man is admitted to the ICU for sepsis secondary to pneumonia and is on multiple vasopressors. The patient has tachycardia. Thyroid function is checked
- TSH 0.2 mcIU/mL (ref: 0.4-4.5)
- Free T4: 0.6 ng/dL (0.7-1.77)
- Total T3 66 ng/dL (80-180)

- What next??

Nonthyroidal Illness Syndrome (Euthyroid Sick Syndrome) ***

- Critically ill patients
- Biologically protective mechanism to prevent excessive tissue catabolism
- Similar pattern to central hypothyroidism





Case 4

- A 50 year old trans woman is referred by their primary care doctor for evaluation of a right sided thyroid nodule. The nodule was seen incidentally on a CT scan of their neck obtained for evaluation of cervical disc disease. You palpate the patient's neck and you don't feel any thyroid nodules or lymphadenopathy.
- What is your next step in management?

Thyroid nodules are super common!

- Some sources estimate that 1 in 6 people has a thyroid nodule that they don’t know about.
- Benign:
  - Multinodular goiter
  - Hashimoto’s thyroiditis
  - Cysts
  - Follicular Adenomas
  - Hurthle Cell Adenomas

Some Thyroid Nodules are Cancer

- Papillary Thyroid Carcinoma – 70-80% (excellent prognosis)
- Follicular Thyroid Carcinoma – 10-20% (excellent prognosis)
- Medullary Thyroid Carcinoma – 1-4% (often good prognosis if caught early)
- Anaplastic Thyroid Carcinoma – <1% (terrible prognosis)

Other:

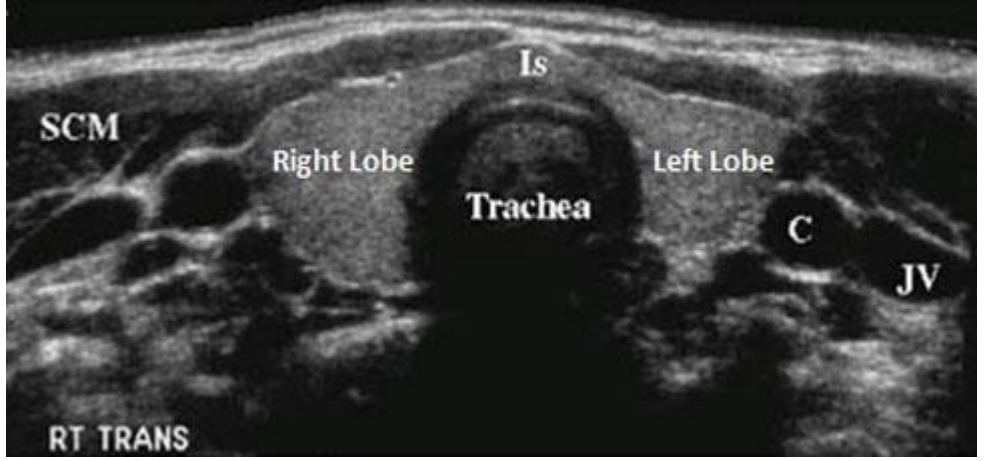
- Primary Thyroid Lymphoma
- Metastatic Carcinoma

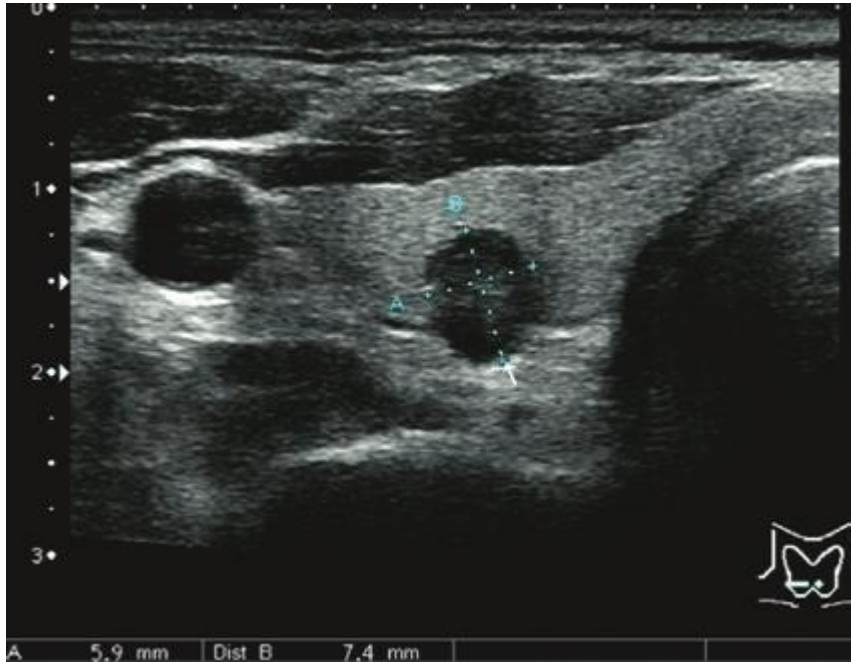
Evaluation

- First Step: TSH
  - If suppressed – could be a toxic adenoma.
    - Then, thyroid uptake and scan
  - Toxic Adenomas are VERY unlikely to carry cancer
- Ultrasound
  - The most sensitive imaging study for assessing thyroid nodules
    - Ultrasonographic features can give us clues if they’re benign or malignant
  - Assess for lymphadenopathy and metastatic spread

Ultrasound

- Suspicious Features:
  - Solid, dense
  - Hypoechoic (dark)
  - Taller than wide (Growing outward)
  - Ill defined margins
  - Calcifications





Fine Needle Aspiration

- Outpatient Procedure – usually <15 minutes
- Ultrasound Guidance
- Performed with or without lidocaine
- 25 or 27 gauge needles used for aspiration
- Look at under microscope

Medullary Thyroid Cancer

- Cancer of C-cells of the thyroid (TBD in the Calcium Metabolism Lecture)
- Association
  - *** Multiple Endocrine Neoplasia Type 2
    - 2A: MTC, Pheochromocytoma, Primary Hyperparathyroidism
    - 2B: MTC, Pheochromocytoma, Mucosal Neuromas, Marfanoid Body Habitus



Thyroid Cancer Management

- Remove the tumor (either half or all of the thyroid, depending on the size of the nodule)
- If you remove the whole thyroid, follow with Radioactive Iodine Ablation – kill off all the remnant thyroid tissue (and remnant thyroid cancer)
- If thyroidectomy + Radioactive Iodine Ablation = monitor for recurrence with serum Thyroglobulin and ultrasound

Q&A

- sanjay.jumani@jefferson.edu